

0.

OI-wiki
Min_25

Min_25

1.

$$f \sum_{i=1}^n f(i) \quad f(i)$$

$$O(n^{\frac{2}{3}}) \quad f(p) \quad p \quad f(p^c) \quad \text{Min_25}$$

$$\forall a, b, f(a)f(b) = f(ab)$$

$$a, b$$

2.

$$G(k, n) : \quad \textcolor{red}{2} \quad n \quad i \quad i, \quad f(i) \quad k$$

$$\sum_{i=2}^n f(i)[L(i) \geq P_k]$$

$$L(i) \quad i \quad P_k \quad k$$

```
int ans=0;
for(int i=2;i<=n;i++){
    if(L[i]>=pri[k]) ans+=f(i);
}
return ans;
```

$$P_k > n \quad G(k, n) = 0$$

3.

$$L(i) \geq P_k \quad f(i) \quad G(k, n) \quad P_k \quad i$$

$$P_j \quad L(i) = P_j$$

$$1. \quad P_j \quad i \quad 1 \quad i \quad \prod_{i=1}^k p_i^{c_i} \quad p_1 = P_j, c_1 = 1$$

$$2. \quad P_j \quad i \quad 1 \quad p_1 = P_j, c_1 > 1$$

$$f(P_j) + \sum_{t=2}^i f(tP_j)[L(t) \geq P_{j+1}]$$

$$\begin{aligned} & i \quad j \quad t = \frac{i}{P_j} \\ & j \quad i \quad t \quad P_j \quad t = 1 \\ & t \neq 1 \quad t \quad P_j \quad P_{j+1} \\ & L(t) \geq P_{j+1} \quad f(i) \quad f(tP_j) \\ & P_j \quad i \quad 1 \quad \gcd(t, P_j) = 1 \quad f \quad f(tP_j) = f(t)f(P_j) \end{aligned}$$

$$f(P_j) + f(P_j) \sum_{t=2}^{\lfloor \frac{n}{P_j} \rfloor} f(t)[L(t) \geq P_{j+1}] \quad G$$

$$f(P_j) + f(P_j)G(j+1, \lfloor \frac{n}{P_j} \rfloor)$$

$$f(P_j), P_j \leq n \quad f(P_j)G(j+1, \lfloor \frac{n}{P_j} \rfloor), \quad P_j^2 \leq n \quad t \geq P_j \quad P_j^2 > n$$

$$t \neq 1 \quad j \quad i \quad t \geq P_j$$

$$\begin{aligned} & i \\ & \sum_{c \geq 2, P_j^c \leq n} (f(P_j^c) + \sum_{t=2}^{\lfloor \frac{n}{P_j^c} \rfloor} f(tP_j^c)[L(t) \geq P_{j+1}]) \quad c \quad P_j \quad i \\ & t = \frac{i}{P_j^c} \\ & j \quad i \quad t \quad P_j \quad t = 1 \\ & t \neq 1 \quad t \quad P_j \quad P_{j+1} \\ & L(t) \geq P_{j+1} \quad f(i) \quad f(tP_j^c) \end{aligned}$$

$$\begin{aligned} & \sum_{c \geq 2, P_j^c \leq n} \left(f(P_j^c) + f(P_j^c) \sum_{t=2}^{\lfloor \frac{n}{P_j^c} \rfloor} f(t)[L(t) \geq P_{j+1}] \right) \\ & G \\ & \sum_{c \geq 2, P_j^c \leq n} \left(f(P_j^c) + f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor) \right) \end{aligned}$$

$$P_j \leq f(i)$$

$$f(P_j) + f(P_j)G(j+1, \lfloor \frac{n}{P_j} \rfloor) + \sum_{c \geq 2, P_j^c \leq n} (f(P_j^c) + f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor))$$

$$k \leq j$$

$$\begin{aligned} G(k, n) &= \left(\sum_{j \geq k, P_j \leq n} \left(f(P_j) + f(P_j)G(j+1, \lfloor \frac{n}{P_j} \rfloor) + \sum_{c \geq 2, P_j^c \leq n} (f(P_j^c) + f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor)) \right) \right) \\ &= \left(\sum_{j \geq k, P_j^2 \leq n} \left(f(P_j)G(j+1, \lfloor \frac{n}{P_j} \rfloor) + \sum_{c \geq 2, P_j^c \leq n} (f(P_j^c) + f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor)) \right) \right) + \sum_{j \geq k, P_j \leq n} f(P_j) \end{aligned}$$

$$\begin{aligned} P_k > n \quad G(k, n) &= 0 \quad \lfloor \sqrt{n} \rfloor < P_j \leq n \quad P_j \quad G(j+1, \lfloor \frac{n}{P_j} \rfloor) = 0 \\ c \geq 2 \quad \lfloor \sqrt{n} \rfloor < P_j \leq n \quad f(P_j) \end{aligned}$$

$$= \left(\sum_{j \geq k, P_j^2 \leq n} \left(\sum_{c \geq 1, P_j^c \leq n} \left(f(P_j^c) + f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor) \right) \right) - f(P_j) \right) + \sum_{j \geq k, P_j \leq n} f(P_j)$$

$$G(k, n) = \left(\sum_{j \geq k, P_j^2 \leq n} \sum_{c \geq 1, P_j^c \leq n} f(P_j^c)([c \neq 1] + G(j+1, \lfloor \frac{n}{P_j^c} \rfloor)) \right) + \sum_{j \geq k, P_j \leq n} f(P_j)$$

$$S(n) = \sum_{p \leq n, p \in \mathbb{P}} f(p)$$

$$G(k, n) = \left(\sum_{j \geq k, P_j^2 \leq n} \sum_{c \geq 1, P_j^c \leq n} f(P_j^c)([c \neq 1] + G(j+1, \lfloor \frac{n}{P_j^c} \rfloor)) \right) + S(n) - S(P_{k-1})$$

$$P_k > n \quad G(k, n) = 0$$

$$\begin{array}{c} \text{OI-wiki} \\ \text{\texttt{OI-wiki}} \end{array}$$

$$\begin{array}{c} c \quad P_j^c \leq n \quad P_j^{c+1} > n \quad \lfloor \frac{n}{P_j^c} \rfloor < P_j < P_{j+1} \\ G(j+1, \lfloor \frac{n}{P_j^c} \rfloor) = 0 \\ f(P_j^c) \\ f(P_j^c)[c \neq 1] \quad c = 1 \\ f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor) \quad c \\ \sum_{c \geq 1, P_j^c \leq n} f(P_j^c)([c \neq 1] + G(j+1, \lfloor \frac{n}{P_j^c} \rfloor)) \\ \sum_{c \geq 1, P_j^{c+1} \leq n} \left(f(P_j^c)G(j+1, \lfloor \frac{n}{P_j^c} \rfloor) + f(P_{j+1}^c) \right) \end{array}$$

$$\sum_{i=1}^n(a_i+b_i)-a_1-b_n=\sum_{i=1}^{n-1}(a_{i+1}+b_i)$$

$$G(k,n)=\left(\sum_{j\geq k,P_j^2\leq n}\sum_{c\geq 1,P_j^{c+1}\leq n}\left(f(P_j^c)G(j+1,\lfloor \frac{n}{P_j^c}\rfloor)+f(P_j^{c+1})\right)\right)+S(n)-S(P_{k-1})$$

$$S \qquad \qquad \text{DP}$$

$$4. \quad S$$

$$\begin{array}{c} S \\ \text{Min}_{p \leq f(p)} \frac{25}{p} \quad f \end{array}$$

$$\begin{array}{c} \sum_{i=1}^r a_i p^{c_i} \\ r \quad c_i \\ S(n) = \sum_{p \leq n, p \in \mathbb{P}} f(p) \\ S(n) = \sum_{p \leq n, p \in \mathbb{P}} \sum_{i=1}^r a_i p^{c_i} = \sum_{i=1}^r a_i \sum_{p \leq n, p \in \mathbb{P}} p^{c_i} \\ p^{c_i} \\ E(k,n) \quad \sum_{i=2}^n [P_k < L(i) \text{ or } i \in \mathbb{P}] i^s \quad k \quad i \quad s \end{array}$$

$$\begin{array}{ccccccc}
& 2 & 3 & 5 & \lfloor \sqrt{n} \rfloor & & \mathbf{n} \\
\text{“ ”} & 2 & & & & & \\
\lfloor \sqrt{n} \rfloor & & & & & & \\
n & A & \lfloor \sqrt{n} \rfloor & & \lfloor \sqrt{n} \rfloor & & \\
P_k, L(i)? & & & & & & \\
L(i) & i & P_k & k & & & \\
& s = c_i & S(n) & E(l, n) & & & \\
P_l & \lfloor \sqrt{n} \rfloor & & & & & \\
& E(k, n) & & & & & \\
& E(k-1, n) & k-1 & i & s & & \\
& k & E(k-1, n) & & & & \\
k & P_k & , & & & &
\end{array}$$

$$\begin{aligned}
&= \sum_{i=2}^{\lfloor \frac{n}{P_k} \rfloor} [P_{k-1} < L(i) \text{ or } i \in \mathbb{P}](iP_k)^s \\
&= P_k^s \sum_{i=2}^{\lfloor \frac{n}{P_k} \rfloor} [P_{k-1} < L(i) \text{ or } i \in \mathbb{P}] i^s \\
&= P_k^s E(k-1, \lfloor \frac{n}{P_k} \rfloor)
\end{aligned}$$

$$\begin{array}{ccccccc}
n & P_k & m & L(\frac{m}{P_k}) & P_k & L(m) & P_k & m & P_k \\
(mP_k)^s & E(k, n) & E(k-1, n) & & & & & & \\
& k & s & m & P_k & P_k & m & &
\end{array}$$

$$P_v P_k \quad v < k$$

$$\begin{aligned}
&\sum_{i=2}^{P_{k-1}} [P_{k-1} < L(i) \text{ or } i \in \mathbb{P}](iP_k)^s \\
P_{k-1} &L(i)
\end{aligned}$$

$$\sum_{i=2}^{P_{k-1}} [i \in \mathbb{P}](iP_k)^s$$

$$P_k^sE(k-1,P_{k-1})$$

$$E(k,n)=E(k-1,n)-P_k^s(E(k-1,\lfloor \frac{n}{P_k}\rfloor)-E(k-1,P_{k-1}))$$

$$P_k>\sqrt{n}\quad k$$

$$E(k,n)=E(k-1,n)-P_k^s[P_k^2\leq n](E(k-1,\lfloor \frac{n}{P_k}\rfloor)-E(k-1,P_{k-1}))$$

$$\begin{array}{l} E(0,x)=\sum_{i=2}^xi^s\qquad 0\qquad 1\\ E\qquad S\\ G\\ \sum_{i=1}^nf(i)=G(1,n)+f(1)=G(1,n)+1\\ \text{Min_25} \end{array}$$

$$5.$$

$$G(k,n)$$

$$G(k,n)=\left(\sum_{j\geq k,P_j^2\leq n}\sum_{c\geq 1,P_j^{c+1}\leq n}\left(f(P_j^c)G(j+1,\lfloor \frac{n}{P_j^c}\rfloor)+f(P_j^{c+1})\right)\right)+S(n)-S(P_{k-1})$$

$$\begin{array}{l} P_j^2\leq n\qquad \Sigma\qquad G(j+1,\lfloor \frac{n}{P_j^c}\rfloor)\qquad G(A,B)\quad P_A\qquad \lfloor \sqrt{n}\rfloor\quad P_{A-1}\\ \lfloor \sqrt{n}\rfloor \end{array}$$

$$\lfloor \frac{\lfloor \frac{n}{x}\rfloor}{y}\rfloor=\lfloor \frac{n}{xy}\rfloor$$

$$B\qquad \lfloor \frac{n}{\lambda}\rfloor$$

$$6$$

$$\begin{array}{c} \forall \lambda \leq n, \lfloor \frac{n}{\lambda} \rfloor \quad 2\sqrt{n} \quad 1, 2, \cdots, \lfloor \sqrt{n} \rfloor, \lfloor \frac{n}{\sqrt{n}-1} \rfloor, \cdots, \lfloor \frac{n}{2} \rfloor, n \\ S \quad P_{A-1}, B \quad S \quad 1, 2, \cdots, \lfloor \sqrt{n} \rfloor, \lfloor \frac{n}{\sqrt{n}-1} \rfloor, \cdots, \lfloor \frac{n}{2} \rfloor, n \\ O(\sqrt{n}) \\ n \quad \texttt{unordered_map} \end{array}$$

$$f$$

$$O(\sum_{i=1}^n f(i)) = O(\int_0^n f(x) \mathrm{d} x)$$

$$\Theta(\sum_{i=1}^n f(i)) = \int_0^n f(x) \mathrm{d} x$$

$$\sigma(n)=\sum_{i=1}^nf(i), \rho(n)=\int_0^nf(x)\mathrm{d} x$$

$$\exists N, c_1, c_2 > 0, \forall n \geq N, n \in \mathbb{N}^+, c_1 \sigma(n) \leq \rho(n) \leq c_2 \sigma(n)$$

$$N=0$$

$$\begin{array}{l} c_1\sigma(n)\leq \rho(n)\leq c_2\sigma(n) \\ \Longleftrightarrow c_1\sum_{i=1}^nf(i)\leq \int_0^nf(x)\mathrm{d} x\leq c_2\sum_{i=1}^nf(i) \end{array}$$

$$\forall i\in\mathbb{N}^+, 1\leq i\leq n\quad u_if(i)\leq \int_{i-1}^if(x)\mathrm{d} x\leq v_if(i)$$

$$\begin{array}{l} c_1\min_{i=1}^nu_ic_2\max_{i=1}^nv_i \\ f\qquad \xi\in [i-1,i]\quad \int_{i-1}^if(x)\mathrm{d} x=f(\xi), \end{array}$$

$$u_i = \frac{f(\xi)}{f(i)} - \varepsilon, v_i = \frac{f(\xi)}{f(i)} + \varepsilon$$

$$\min_{25} \quad O\left(\frac{n^{\frac{3}{4}}}{\log n}\right) \quad O(n^{1-\varepsilon}) \quad \text{OI-wiki}$$

6.

```
#include <bits/stdc++.h>
#define sor(i, l, r) for (int i = l; i <= r; i++)
#define int __int128
using namespace std;
/* */
namespace Revitalize
{
    const int N = 2e6+7, P = 1e9 + 7;
    int n,m;
    int Ex01[N], Ex02[N];
    int ids[N], sqrs[N];

    bool vis[N], cuur;
    int pri[N], Cl[N], I[N], II[N];
    inline void LinearSieve(int lim)
    {
        ids[1] = sqrs[1] = 1;
        for (int i = 2; i <= lim; i++)
        {
            if (!vis[i])
            {
                pri[++pri[0]] = i;
                ids[pri[0]] = (ids[pri[0] - 1] + i) % P;
                sqrs[pri[0]] = (sqrs[pri[0] - 1] + i * i) % P;
            }
            for (int j = 1; j <= pri[0] && i * pri[j] <= lim; j++)
            {
                vis[i * pri[j]] = 1;
                if (i % pri[j] == 0)
                {
                    break;
                }
            }
        }
    }

    inline int L(int x){
        return (x <= sqrt((double)n)) ? (I[x]) : (II[n/x]);
    }
}
```

```

}
inline void Ex(int n)
{
    for (int k = 1; pri[k] * pri[k] <= n; k++)
    {
        for (int i = 1; i <= m && pri[k] * pri[k] <= Cl[i]; i++)
        {
            Ex01[i] = (Ex01[i] - pri[k] * (Ex01[L(Cl[i] / pri[k])] - ids[k - 1]) % P + P) % P;
            Ex02[i] = (Ex02[i] - pri[k] * pri[k] % P * (Ex02[L(Cl[i] / pri[k])] - sqrs[k - 1]) % P + P) % P;
        }
    }
}
int G(int k, int n)
{
    if (pri[k] > n)
    {
        return 0;
    }
    int ans = 0;
    for (int j = k; pri[j] * pri[j] <= n; j++)
    {
        for (int Pjc = pri[j]; Pjc * pri[j] <= n; Pjc *= pri[j])
        {
            ans += Pjc % P * (Pjc - 1 + P) % P * G(j + 1, n / Pjc) % P;
            ans %= P;
            ans += (Pjc % P * pri[j] % P) * (Pjc * pri[j] % P - 1 + P) % P;
            ans %= P;
        }
    }
    ans += (Ex02[L(n)] - Ex01[L(n)] + P) % P;
    ans -= (sqrs[k - 1] - ids[k - 1] + P) % P;
    return ans;
}
inline void Lx(int n)
{
    int l = 1, r = 0;
    while (l <= n) {
        r = n / (n / l);
        Cl[++m] = n / l;
        Ex01[m] = Cl[m] * (Cl[m] + 1) / 2 % P - 1;
        Ex02[m] = Cl[m] * (Cl[m] + 1) % P * (Cl[m] + Cl[m] + 1) % P * 166666668 % P - 1;
        if (Cl[m] <= sqrt((double)(n)))
            I[Cl[m]] = m;
        else
            II[n / Cl[m]] = m;
        l = r + 1;
    }
}

```

```

    }
}
inline void work()
{
    read(n);
    LinearSieve((int)(1000000));
    Lx(n);
    Ex(n);
    int ans = G(1, n) % P + 1;
    print(ans);
}
}
signed main()
{
    ios::sync_with_stdio(false);
    cin.tie(0), cout.tie(0);
    return Revitalize::work(), 0;
}

__int128

```

7.

- Min_26
- Min_25
- zhoukangyang $O(\sqrt{n}\text{polylog}(n))$

Inf.

luuia
Collapsarr

CWH